Bits-Back Coding: Why the ELBO is Exactly Right

1. Sample $z \sim q_\phi(z|x)$

Consume bits from shared coder state

$-\log q_\phi(z|x)$ bits

2. Send z using prior

Entropy coding with $p(z)$

$-\log p(z)$ bits

3. Send x given z

Entropy coding with $p(x|z)$

$-\log p(x|z)$ bits

4. Reclaim bits

Decode under $q_\phi(z|x)$ to recover bits

$+\log q_\phi(z|x)$ bits back

Net cost: $-\log p(z) - \log p(x|z) + \log q_\phi(z|x)$
$= -\log p(x, z) + \log q_\phi(z|x) = -\text{ELBO}$